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SUPERSEDING
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September 9, 1968

FEDERAL SPECIFICATION

PIPE, CORRUGATED (IRON OR STEEL, ZINC COATED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers zinc coated corrugated iron or steel pipe for culverts, underdrains, conduits, and miscellaneous applications.

1.2 Classification.

1.2.1 Class, series, and shape. Pipe, pipe arches, and arches shall be of the following classes, series (see table II), and shapes, as specified (see 6.2):

Class I - Pipe, annular corrugations, series A and B.
II - Pipe, helical corrugations, series A, B, and C.
III - Structural plate pipe, annular corrugations, series D, bolted field assembly.

Shape 1 - Pipe, full circular cross section.
2 - Pipe, factory elongated.
3 - Pipe arch cross section.
4 - Arches of circular arc cross section.

1.2.2 Coating. The following coatings, in addition to the zinc coat, are available where additional protection is required (see 6.2):

Coating A - Fully bituminous coated.
B - Half bituminous coated, paved invert.
C - Fully bituminous coated, paved invert.
D - Paved invert only.
E - Fully bituminous coated, fully paved.
F - Field applied asphalt mastic coating.
G - Asbestos-impregnated, bituminous coated.
H - Asbestos-impregnated, bituminous coated, paved invert.
J - Asbestos-impregnated, bituminous coated, fully paved.
P - Mill applied pitch resin or polymer coatings.

Type A - Sheet coated with pitch resin or polymer in a thickness of 0.010 inch minimum one surface only.
Type B - Sheet coated with pitch resin or polymer in a thickness of 0.010 inch minimum on one surface and polymer in a thickness of 0.0003 inch minimum on the other side.
Type C - Sheet coated with pitch resin or polymer in a thickness of 0.010 inch minimum on both sides.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

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Federal Specifications:

QQ-S-781 - Strapping, Steel, Flat and Seals.
TT-P-641 - Primer Coating, Zinc Dust-Zinc Oxide (for Galvanized Surfaces).
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Box, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing.
Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-21035 - Paint, High Zinc Dust Content, Galvanizing Repair.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

A 31 - Specification for Boiler Rivet Steel and Rivets.
A 36 - Specification for Structural Steel.
A 90 - Method of Test for Weight of Coating on Zinc Coated (Galvanized) Iron or Steel Articles.
A 153 - Specification for Zinc-Coated (Hot-Dip) on Iron and Steel Hardware.
A 164 - Specification for Electrodeposited Zinc on Steel.
A 283 - Specification for Low and Intermediate Tensile Strength Carbon Steel Plates of Structural Quality.
A 307 - Specification for Low-Carbon Steel Externally and Internally Threaded Standard Fasteners.
A 325 - Specification for High Strength Carbon Steel Bolts for Structural Joints, including Suitable Nuts and Plain Hardened Washers.
A 370 - Methods and Definitions for Mechanical Testing of Steel Products.
A 444 - Specification for Zinc-Coated (Galvanized) Iron or Steel Sheets for Culverts and Underdrains.
A 449 - Specification for Quenched and Tempered Steel Bolts and Studs.
A 563 - Specification for Carbon Steel Nuts.
A 569 - Specification for Sheet Carbon (0.15 Maximum Percent) Hot Rolled and Strip, Commercial Quality.

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American Society for Testing and Materials (ASTM) Standards (con.):

D 93 - Method of Test for Flash Point by Pensky Martens Closed Tester.
D 147 - Bituminous Mastics, Grouts, and Like Mixtures, Testing.
D 149 - Method of Tests for Dielectric Breakdown Voltage and Dielectric Strength of Electrical insulating Materials at Commercial Power Frequencies.
D 529 - Recommended Practice for Accelerated Weathering Test of Bituminous Materials.
D 543 - Method of Test for Resistance of Plastics to Chemical Reagents.
D 568 - Method of Test for Flammability of Flexible Plastics.
D 570 - Method of Test for Water Absorption of Plastics.
D 638 - Method of Test for Tensile Properties of Plastics.
D 658 - Method of Test for Abrasion Resistance of Coatings of Paint, Varnish, Lacquer, and Related Products with the Blast Abrasion Tester.
D 903 - Method of Test for Peel or Stripping Strength of Adhesive Bonds.
D 1004 - Method of Test for Tear Resistance of Plastic Film and Sheeting.
D 1005 - Dry Film Thickness of Organic Coatings, Measurement of.
D 1010 - Method of Test for Asphalt Emulsions for Use as Protective Coatings for Metal.
D 1208 - Method of Test for Common Properties for Some Pigments.
D 1856 - Method of est for Recovery of Asphalt from Solution by Abson Method.
G 22 - Recommended Practice for Determining Resistance of Plastics to Bacteria.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American Association of State Highway Officials (AASHO) Publications:

M 167 - Specification for Structural Plate for Pipe, Pipe-Arches, and Arches.
M 190 - Specification for Bituminous Coated Riveted Corrugated Metal Culvert Pipe and Pipe Arches.

(Application for copies should be addressed to the American Association of State Highway Officials, 341 National Press Building, Washington, DC 20004.)

American National Standards Institute (ANSI) Standards:
B18.2.2 - Square and Hex Nuts.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza., Chicago, IL 60606.)

3 REQUIREMENTS

3.1 Classes I and II, sheet material.

3.1.1 Material. All materials and tests shall conform to ASTM A444. In addition, the mechanical properties of the galvanized sheets for the flat material prior to corrugating shall be as follows (see 4.4.1):

Minimum tensile strength, p.s.i.	Minimum yield strength, p.s.i.	Minimum elongation, in 2 inches, percent
45,000	33,000	20

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3.1.2 Perforations for drainage. When specified (see 6.2), perforations shall be arranged in two groups or longitudinal rows, placed symmetrically with one group on each side of a nonperforated bottom segment corresponding to the flow line of the pipe. Within each group the rows of perforations shall be located in the inside crests or on the neutral axis of all corrugations, except that perforations are not required within 6 inches of each end of each length of pipe or in the crests of corrugations where seams are located. The perforations shall have a diameter of approximately 3/8 inch. When bituminous coated pipe is specified the clear diameter of each perforation after coating shall be not less than 75 percent of the diameter of perforation before coating. The minimum number of longitudinal rows of perforations and the minimum width of the nonperforated bottom segment shall be as shown in table I.

TABLE I. Subdrain pipe perforation, shape 1 only 1/

Nominal internal diameter	Class of pipe	Number of longitudinal rows of perforations in pipe (2 groups)	Width of nonperforated bottom segment
Inches		Minimum	Minimum, inches
6	II	4	4-1/2
8	I, II	4	7
10	I, II	4	9
12	I, II	6	9-1/2
15	I, II	6	13
18	I, II	6	16-1/2
21	I, II	6	20
24	I, II	8	22

1/ Other types and patterns of perforations may be furnished, when specified, (see 6.2).

3.2 Classes I, II, and III.

3.2.1 Corrugations. Corrugations shall form smooth continuous curves and tangents. The radii of curvature of the corrugations shall be at least one-half the depth of corrugation. The crests and valleys of annular corrugations shall form circumferential rings (complete or partial) about the axis of the pipe. Helical corrugations shall form helices about the axis and the direction of crests and valleys shall be not less than 50° from the longitudinal axis of the pipe. The series identification and dimensions of corrugations for three classes of pipe shall be in accordance with table II. The depth of corrugations shall not underrun those specified by more than 5 percent.

TABLE II. Series designations, corrugation dimensions in inches

Class	Series	Diameter range	Depth	Pitch		Specification
				Minimum	Maximum	
I annular	A	8 to 96	1/2	2-1/4	2-3/4	ASTM A444
	B	36 to 120	1	2-3/4	3-1/4	
II helical	A	12 to 96	1/2	1-7/8	2-3/4	
	B	36 to 120	1	2-3/4	3-1/4	
	C	6, 8, 10	1/4	1-3/8	1-7/8	
III annular	D	60 to 252	2	5-3/4	6-1/4	AASHO M 167

3.2.2 Welds, miscellaneous. When required, miscellaneous welding, except that covered in 3.3.3 and 3.4.2, shall be in accordance with drawings and specifications that are supplied by the manufacturer and approved by the contracting officer. After completion of such welding, all welds and both surfaces of adjoining metal, adjacent to the welds, shall be thoroughly wire brushed and painted with at least two coats of zinc dust-zinc oxide primer conforming to TT-P-641, type III or MIL-P-21035.

3.2.3 Thickness of zinc-coated sheet and plate. Unless otherwise specified (see 6.2), thickness and tolerance on thickness shall be in accordance with table III.

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TABLE III. Thickness and tolerance, zinc-coated sheet and plate

Specified thickness, inches	Tolerance under, inches 1/	Nominal gage 2/
0.052	0.006	18
.064	.007	16
.079	.007	14
.109	.012	12
.138	.012	10
.168	.012	8
.188	.012	7
.218	.012	5
.249	.012	3
.280	.012	1

1/ No limit on over thickness.
2/ For informational purposes only.

3.2.4 Weight of zinc coat. Weight of the zinc coat shall be as specified in ASTM A444 for classes I and II and shall be applied at the rate of 3 ounces per square foot (see 3.5.2) for class III (see 4.4.2.1 and 4.4.2.2).

3.2.5 Length, diameter, and number. Number of sections, length, and diameter of sections, and total length required shall be as specified (see 6.2).

3.3 Class I.

3.3.1 Fabrication. Pipe shall be series A or B, of riveted or resistance spot welded lap joint construction as specified (see 6.2). For series B pipe, 1/2 inch diameter bolts and nuts meeting the requirements of ASTM A 449 and ASTM A 563, grade C, respectively, and galvanized in conformance with ASTM A 153 may be used in lieu of rivets or spot welds on a one for one replacement ratio. Bolts heads and nuts shall be shaped to provide adequate bearing or standard type bolts and nuts with specially shaped washers shall be used in lieu thereof. Washers shall be galvanized in accordance with ASTM A 153 and shall conform to the requirements for washers, except dimensionally, of ASTM A 325.

3.3.2 Rivets and riveted construction.

3.3.2.1 Rivets. All rivets shall conform to ASTM A 31, grade A, and shall be zinc coated by hot dipping in accordance with ASTM A 153 or electroplated in accordance with ASTM A 164, type RS. Rivets for class I, series A pipe, shall be not less than 5/16 inch in diameter for metal 0.079 inch thick (14 gage) or lighter and not less than 3/8 inch in diameter for metal 0.109 inch thick (12 gage) or heavier. Rivets for class I, series B pipe, shall be not less than 3/8 inch in diameter for metal 0.079 inch thick or lighter and not less than 7/16 inch in diameter for metal 0.109 inch thick or heavier. All rivets shall be driven cold in such a manner that the sheets shall be drawn tightly together throughout the entire lap. The center of all rivets shall be at least twice their diameter from the edges of the sheets. All rivets shall have a neat, workmanlike, and full head of a form acceptable to the contracting officer, shall be driven with heads concentric with shanks, without bending, and shall completely fill the holes.

3.3.2.2 Rivets for longitudinal and circumferential seams. Longitudinal seams shall be riveted with at least one rivet in the valley of each corrugation. The longitudinal seams of class I, series A pipe 42 inches or more in diameter and all diameters of series B pipe shall have two rivets in the valley of each corrugation and the nominal distance between the centerline of the two rows of rivets shall be three times the diameter of the rivets. Circumferential seams shall have a rivet spacing not greater than 6 inches except that 6 rivets will be sufficient for 12-inch diameter pipe. When 1/2 inch round ASTM A 449 bolts are used in lieu of rivets for series B pipe, they shall be tightened so that the sheets shall be drawn tightly together throughout the entire lap but to not more than 90 foot-pounds torque.

3.3.3 Resistance spot welded fabrication.

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3.3.3.1 Spacing, number, and dimensions. Resistance spot welds shall be made in such a manner that the metal shall be drawn tightly together throughout the lap. The center of a spot weld shall be not less than its radius plus 1/4 inch from the edge of the metal sheets. The spot welds in longitudinal and circumferential seams shall be located in the valley, crest, or tangent of the corrugation. Longitudinal seams of spot welded series A pipe 42 inches or more in diameter, and of all series B pipe shall have two spot welds in each corrugation and the two rows of welds shall be not closer center to center than two weld diameters. Circumferential seams shall have a spot weld spacing not greater than 6 inches except that 6 welds will be sufficient for 12-inch diameter pipe.

3.3.3.2 Quality and procedure approval. The welding shall be performed in such a manner that (a) the exterior surfaces of 90 percent or more of the spot welds on a length of pipe show no evidence of melting or burning of base metal, and (b) base metal shall not be exposed when the area adjacent to the electrode contact surface area is wire brushed. Discoloration of spot weld surfaces shall not be cause for rejection. Before welding of pipe is permitted each welding machine and operator shall be qualified by the tests specified in 4.4.4.

3.3.3.3 Shear strength. Single shear strength of spot welds shall be in accordance with table IV.

TABLE IV. Shear strength, spot welds

Corrugated sheet, specified thickness	Single shear strength, minimum
Inches	Pounds
0.064	4,000
.079	5,000
.109	7,000
.138	8,000
.168	10,000

3.3.4 Coupling bands. Coupling bands for corrugated steel pipe shall provide circumferential and longitudinal strength to preserve the pipe alignment, prevent separation of pipe and minimize infiltration of fill material. Bands shall have corrugations or projections that mesh with the pipe corrugations. Coupling bands may be the next thickness lighter than that used for the pipe but not more than 0.109 nor less than 0.052 inch thick.

3.3.4.1 Bands with corrugations. The bands shall be formed to fit the pipe shape specified and shall be not less than 7 inches wide for pipe 6 to 30 inches in diameter, 12 inches wide for pipe 36 inches to 60 inches in diameter, and 24 inches wide for pipe 66 to 120 inches in diameter. Bands shall be either two-piece or one-piece unless otherwise specified (see 6.2). The bands shall be so constructed as to lap on an equal portion of the pipe sections to be connected. The bands shall have end connection angles, conforming to ASTM A 36, zinc coated in accordance with ASTM A 153, not less than 2 inches by 2 inches by 3/16 inch by the width of the band minus 1 inch, adequately fastened to each end of band and half bands. The bands shall be of such length as to hold the two connected sections of pipe rigidly together when tightened in place by the connection bolts furnished. The 7-inch bands shall have at least two zinc coated connection bolts, conforming to ASTM A 307, grade A, electroplated in accordance with ASTM A 164, type RS, not less than 1/2 inch in diameter per connection. The 12 inch bands shall have not less than three and the 24-inch bands not less than five such bolts per connection.

3.3.4.2 Bands with projections. The projections shall conform substantially to the shape and depth of the pipe corrugations and shall be in circumferential rows with no less than seven projections per row. The bands may be used on annular or helical corrugated pipe or a combination of both and shall meet the requirements or table V. Requirements for bolts and nuts shall be in accordance with 3.3.4.1.

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TABLE V. Pipe coupling band requirements (with projections)

Diameters	Widths of bands		Total rows of projections and series		Sizes of connecting angles and series		Number of 1/2 inch diameter bolts
Inches	Min.	Inches	Min.	Series	Inches	Series	Min.
12 to 54	10-1/2		2	A	2 by 2 by 3/16	A	2
60 to 96	16-1/4		4	A	2 by 2 by 3/16	A	3
36 to 54	12		2	B	2 by 2 by 3/16	B	2
60 to 120	18		4	B	2 by 2 by 3/16	B	3

3.3.4.3 Other fastening bands. Other equally effective fastening bands or methods for connecting pipe sections may be used if approved by the contracting officer.

3.3.5 Pipe diameters. The average inside diameter of circular pipe shall not vary more than one percent of 1/2 inch, whichever is greater, from the specified nominal diameter when measured on the inside crest of the corrugations.

3.3.6 Shapes, class I.

3.3.6.1 Shape 1. Pipe furnished as shape 1 shall be of full circle shape.

3.3.6.2 Shape 2. Pipe furnished as shape 2 shall be fabricated by reforming circular pipe to an approximate ellipse with the vertical diameter approximately 5 percent greater than the diameter of a corresponding circular pipe.

3.3.6.3 Shape 3. This shape shall be fabricated by reforming a circular pipe to a multi-centered pipe having an arch shaped top with a slightly convex curved integral bottom. The lapped longitudinal seams shall be staggered so as to alternate on each side of the center of arch top. Alternatively, the shape may be fabricated by curving individual sheets to form circular arcs as required for the shape specified. The pipe shall conform to the requirements of Table VI. Span and rise required shall be as specified (see 6.2).

TABLE VI. Dimensions, classes I and II pipe arches, inches 1/, 4/

Span 2/	Rise 3/	Diameter equivalent round pipe	Corner radius (minimum)
Shape 3 - Series A, 2-2/3 by 1/2 inch and 2 by 1/2 inch			
18	11	15	3
22	13	18	3
25	16	21	3
29	18	24	3
36	22	30	3
43	27	36	3-1/2
50	31	42	4
58	36	48	5
65	40	54	6
72	44	60	7
79	49	66	8
85	54	72	9
Shape 3 - Series B, 3 by 1 inch			
43	27	36	5
50	31	42	6
58	36	48	7
65	40	54	8
72	44	60	9
73	55	66	12
81	59	72	14
87	63	78	14
95	67	84	16
103	71	90	16
112	71	96	18
117	79	102	18
128	83	108	18
137	87	114	18
142	91	120	18

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NOTE 1/ All dimensions are measured from the inside crest of corrugations, and the shape shall be essentially of the pipe arch configuration.
2/ Span: Minus tolerance, 5 percent of equivalent pipe diameter but not less than 2 inches.
3/ Rise: Plus tolerance, 5 percent of equivalent pipe diameter but not less than 2 inches.
4/ Ends of joining sections shall be of the same geometry and shall be adaptable to the coupling bands specified in 3.3.4.

3.4 Class II.

3.4.1 Pipe fabrication. Pipe furnished under this class shall be series A, series B, or series C fabricated with helical corrugations as specified in 3.3 and shall be in shape 1, shape 2, or shape 3 as specified (see 6.2). Perforations when specified shall be in accordance with 3.1.2. Smooth lined pipe when specified (see 6.2), shall be fabricated with a smooth liner and helically corrugated series A or series B shell integrally attached at helical lock seams not more than 30 inches apart and extending from end to end length of pipe.

3.4.2 Seams. Pipe with helical corrugations shall have a continuous seam extending from end to end of each length of pipe section. Helical seams shall be fabricated in such a manner that they will develop the full strength of the pipe and not affect shape or nominal diameter of the pipe. Folded lock seams or ultra-high frequency resistance butt-welded seams shall be used. Folded lock seams shall be formed with sufficient pressure to prevent any seam slippage that would appreciably affect the load carrying capacity of the pipe but without damaging the metal to such an extent that a plane of weakness is created. The metal used in fabricating lock seams shall be one that will permit cold forming without such damage. Welded seams shall be controlled such that the combined width of the weld and adjacent spelter coating burned by welding does not exceed three times the metal thickness. Damage outside this width shall be repaired as required in 3.2.2. The contractor shall certify that the welds have been tested and found satisfactory. Such tests shall be made at the beginning of each shift and also when a change of metal thickness is made. The tests shall be acceptable to the contracting officer.

3.4.3 Coupling bands. Bands shall have corrugations that mesh with the pipe corrugations or shall have projections that mesh or engage with the pipe corrugations. The projections shall conform substantially to the shape and depth of the pipe corrugations and shall be in circumferential rows with one projection in each row for each corrugation of the helical pipe sections connected. The bands with projections shall conform to table V.

3.4.3.1 Length, gage, and fastening details. The lengths, gage, and fastening details for bands shall be as specified in 3.3.4 except that integral formed flanges may be used instead of connections angles for pipe diameters up to 24 inches inclusive and that 0.052 inch bands may be used with 0.052 inch pipe wall.

3.4.3.2 Other fastening bands. Other equally effective fastening bands or methods for connecting the pipe sections may be used if approved by the contracting officer (see 6.2).

3.4.4 Pipe diameters. The average inside diameter of circular pipe shall not vary more than one percent or 1/2 inch, whichever is greater, from the specified nominal diameter when measured on the inside crests of the corrugations.

3.4.5 Shapes, class II.

3.4.5.1 Shape 1. Pipe furnished as shape 1 shall be of full circle shape.

3.4.5.2 Shape 2. Pipe furnished as shape 2 shall be fabricated to form an approximate ellipse with the vertical diameter approximately 5 percent greater than the diameter of a corresponding circular pipe.

3.4.5.3 Shape 3. This shape will be fabricated by reforming a circular pipe to multicentered pipe having an arch shaped top with a slightly convex curved integral bottom. The pipe shall conform to the requirements of table VI. Span and rise required shall be in accordance with 3.3.6.3, table VI, as specified (see 6.2).

3.5 Class III.

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3.5.1 Material. All materials and tests shall conform to AASHO M 167. In addition, the mechanical properties of the black sheets or plates for the flat material prior to corrugating shall be as follows (see 4.4.1):

Minimum tensile strength, p.s.i.	Minimum yield strength, p.s.i.	Minimum elongation in 2 inches, percent
42,000	28,000	30

3.5.2 Zinc coating. Pipe shall be zinc coated in accordance with AASHO M 167 except it shall be applied at the rate of 3.0 ounces per square foot of double exposed surface (see 3.2.4 and 4.4.2.2). The average spelter coating from the required samples, or any one specimen shall not be less than 2.7 ounces per square foot.

3.5.3 Bolts. All bolts and nuts shall be 3/4 inch diameter meeting the provisions of ASTM A 449 and ASTM A 563, grade C, respectively, and shall be hot dip galvanized in accordance with the requirements of ASTM A 153. Bolts and nuts shall conform to the dimensions of ANSI B18.2.2 except that the bearing faces of bolts and nuts shall be symmetrically shaped to a one inch radius spherical surface with the maximum bolt head and nut thickness not less than the minimum specified for regular hexagon bolts and nuts. Regular hexagon bolts and nuts with hot dipped galvanized washers to fit the pipe corrugations may be used in lieu of the spherically faced bolt heads and nuts. The number of bolts of each size and length furnished shall be 5 percent plus 2 in excess of the theoretical number required to field erect the structure or structures. Bolt lengths shall be such as to result in at least "full nuts" when tightened in place. There shall be at least four bolts per foot of longitudinal seam. If specified, 6 or 8 bolts per foot of longitudinal seam shall be furnished with plates 0.28 inch thick (see 4.4.3 and 6.2).

3.5.4 Structural plates. Structural plates furnished under this class shall be fabricated with bolted lap seams. Plates shall consist of structural units of galvanized corrugated metal formed to provide lap seams. Single plates shall be furnished in standard sizes to permit structure length increments of two feet. The plates at longitudinal and circumferential seams shall be fabricated for connection by bolts as specified in 3.5.3. Joints shall be staggered so that not more than three plates come together at any one point. Each plate shall be curved to form one or more circular arcs as required for the shape specified.

3.5.4.1 Bolt holes. The bolt holes shall be punched so that all plates having like dimensions, curvature, and same number of bolts per foot of seam shall be interchangeable. Each plate shall be curved to the proper radius so that cross-sectional dimensions of the finished structure will be as indicated on the drawings or as required for the type specified. Bolt holes along the edge of the plates that will form longitudinal seams in the finished structure shall be staggered in rows two inches apart, with holes in one row in the valleys and holes in the other row in the crest of the corrugations. Bolt holes along those edges of the plates that will form the circumferential seams in the finished structure shall provide for a bolt spacing of not more than 10 inches. The minimum distance from the center of the hole to the edge of the plate shall not be less than 1-3/4 times the diameter of the bolt. Holes shall be furnished as required, for connecting to the end and opening stiffeners, headwall anchors, bearings, and miscellaneous attachments. The diameter of the bolt holes in the longitudinal seams shall not exceed the bolt diameter by more than 1/8 inch except those at the plate corners and in circumferential seams. Diameter of holes in circumferential seams, including corner holes, shall not exceed the bolt diameter by more than 1/2 inch and the average of the diameters on the major and minor axis shall not exceed the bolt diameter by more than 1/4 inch.

3.5.4.2 Edge of plate and identification. Plates for forming skewed or beveled ends of structures shall be accurately cut to form to prescribed final shape. Cut edges of plates shall be free from notches, gouges, oxide, or burrs and shall present a workmanlike finish. Legible indentification numerals shall be placed on each part plate to designate its proper position in the finished structure as indicated on approved erection drawings.

3.5.5 Shapes, class III.

3.5.5.1 Shape 1. Structural plates furnished under shape 1 shall form a full circle when erected.

3.5.5.2 Shape 2. Structural plates furnished under this type shall be erected with the longest diameter vertical. The longest diameter shall be approximately 5 percent greater than the diameter of a corresponding circular pipe.

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3.5.5.3 Shape 3. Structural plates for pipe arches shall form cross sections made up of four circular arcs tangent to each other at their junction and symmetrical about the vertical axis. Dimensions shall conform to the procurement plans.

3.5.5.4 Shape 4.

3.5.5.4.1 Shape and corrugations. Structural plates for arches shall be curved to form an arch of circular arc cross section with corrugations at right angles to the axis of the structure. Dimensions, including span and rise required, shall be as shown on the procurement plans, as specified (see 6.2).

3.5.5.4.2 Bearings for arches. When specified (see 6.2), metal bearings for arches shall be either rolled structural angles or channels, conforming to ASTM A 36, A 283, grade C, or A 569, not less than 3/16 inch in thickness, galvanized in accordance with ASTM A 153. Anchorage of bearings to foundation shall be fabricated as shown on the procurement plans, with spacing on not more than 24 inches on centers. Bolt holes shall be punched at not more than 24 inches on center in the vertical leg of bearings to match corresponding bolt holes in the bottom holes in the bottom arch plate. Bolts for connection arch plates to bearings shall conform to 3.5.3.

3.5.6 Identification marking. In addition to the idenfitication required by AASHO M 167, each class III plate shall be stamped with its coating lot number. The brand shall be removed or obliterated by the fabricator on all material in a lot or heat when control tests, as outlined herein, do not show conformance to this specification.

3.6 Coatings, bituminous. Coating A to J, inclusive, except for coating F (see 3.6.6), shall be an asphalt conforming to section 4 of AASHO M 190 (see 4.4.5.1).

3.6.1 Coating A. The pipe shall be uniformly ocated with bituminous material inside and outside to a minimum thickness of 0.05 of an inch measured on the crest of the corrugations (see 4.4.5.2).

3.6.2 Coating B. The pipe shall be coated for approximately the lower one-half of the circumference (bottom of pipe installed) inside and outside to a minimum thickness of 0.05 of an inch and in addition, the bituminous material shall be so applied as to form a smooth pavement in the invert (inside bottom of pipe when installed) filling the corrugations for at least 25 percent of the circumference for shapes 1 and 2 and 40 percent for shape 3. The invert pavement shall have a minimum thickness of one-eight (1/8) inch above the crest of the corrugations except where the upper edges thereof intersect the corrugations (see 4.4.5.2).

3.6.3 Coating C. The pipe shall be full coated as specified in 3.6.1 and in addition a smooth invert pavement shall be provided as specified in 3.6.2.

3.6.4 Coating D. A smooth invert pavement only shall be provided as specified in 3.6.2.

3.6.5 Coating E. The pipe shall be bituminous coated as specified in 3.6.1 and in addition to the bituminous material shall be so applied as to form a smooth pavement over the complete inner surface of the pipe. This pavement shall have a minimum thickness of 1/8 inch over the crest of corrugations (see 4.4.5).

3.6.6 Coating F. Field applied asphalt mastic coating.

3.6.6.1 Manufacture. The component ingredients of the mastic shall be mixed and dispersed as required to produce a mastic of light plastic consistency that is free of added water, coal tar, or coal tar solvents, and shall be processed in accordance with high grade commercial practice to produce a uniform product suitable for the purpose intended. The final product shall be capable of being broken up with a paddle to a smooth uniform consistency.

3.6.6.2 Quantitative requirements. The mastic shall conform to the requirements in table VII (see 4.4.5.3.1, and 4.4.5.3.3 to 4.4.5.3.7).

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TABLE VII. Quantitative requirements field applied mastic coating, F

Characteristics	Requirements	
	Minimum	Maximum
Filler		
Percent by weight	20.0	40.0
Inorganic filler containing short fibre asbestos and fine inert minerals, percent	100	-
Vehicles, percent	60	-
Asphalts, percent	45	65
Petroleum, percent	35	55
Nonvolatile bitumens, calculated by difference, percent	30.0	-
Pigment, percent by volume of nonvolatile	25.0	-
Weight per gallon, pounds	8.0	9.5
Flash point of solvents, °F.	80.0	-

3.6.6.3 Drying time requirement. A wet film of coating, sufficient to deposit a 1/16 inch thick dry film, shall set to touch in 4 hours, and shall reach practical hardness in 36 hours (see 4.4.5.3.8).

3.6.6.4 Color. The color shall be black.

3.6.6.5 Working properties. The mastic shall be capable of applicable by brush and trowel in addition to spraying with standard high pressure spray equipment of the types recommended for application of mastics. The coating shall be uniform continous, and free from any pin holing when tested in accordance with 4.4.5.3.9.

3.6.6.6 Condition in container. The mastic shall show little or no settling in a freshly opened full container; shall show no curdling, livering, vehicle separation, or caking, and shall be free from lumps or skins even after a period of storage of 12 months from the date of delivery in a full, tightly covered container stored at a temperature of 70° to 90°F.

3.6.6.7 Sag. A 1/8 inch wet film shall not sag or pile up on the lower half of the panel when tested in accordance with 4.4.5.3.10.

3.6.6.8 Ingredients.

3.6.6.8.1 Asphalts. Bitumens used shall consist of either or both naturally occuring or petroleum derived asphalts and may include up to 50 percent by weight of gilsonite; any petroleum asphalt used shall be a straight run, steam, or vacuum refined residue and shall be air blown in accordance with good commercial practice. The melting point of the asphalt when tested in accordance with 4.4.5.3.2 shall be not less than 170°F. and the penetration at 77°F. shall be 10 to 20.

3.6.6.8.2 Fillers. Fillers used shall be suitable inorganic minerals and short fibre asbestos. Sand and other harsh abrasive materials shall not be used. The fillers (exclusive of asbestos) shall be fine enough to pass through a 325 mesh screen. Not more than 20 percent by weight of the total mastic shall be retained on a 325 mesh screen as determined by washing the mastic through the screen with a suitable solvent (see 4.4.5.3.1).

3.6.6.8.3 Petroleum solvents. The volatile solvents shall consist of any suitable asphaltic petroleum solvents.

3.6.6.9 Marking of containers. Each container shall be suitably labeled or marked with the following information:

(a) NAME: Cold Applied Asphalt Mastic
SPECIFICATION:
COLOR: Black
LOT NUMBER:
DATE OF MANUFACTURE:
QUANTITY OF PAINT IN CONTAINER:
MANUFACTURER'S NAME AND ADDRESS:

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(b) Directions for use

The mastic shall be applied to a clean surface free of oil grease moisture with the atmosphere temperature at least 40°F. and the humidity low enough that the surface of the metal can be kept dry.

The mastic shall be mixed thoroughly before using. It shall not be thinned unless absolutely necessary. Add only mineral spirits for thinning. Apply by spray, use high pressure spray equipment. If applied by brush, a daubing action shall be used. Mastic may also be applied by trowel. Apply at a uniform rate of 4 to 6 gallons per 100 square feet.

3.6.7 Coating G. This type of pipe shall be fabricated from asbestos-impregnated sheets, the base metal of which shall conform to 3.1.1. Both sides of the metal sheets shall be coated with a layer of asbestos fibre applied by pressing sheet asbestos fibre into the molten zinc bonding medium. Immediately after solidification of the zinc the asbestos fibre shall be thoroughly impregnated with a bituminous saturant. The finished sheets shall be of first class commercial quality free from the amount and quality of zinc, asbestos fibres, or bituminous saturant for coatings G, H, and J. The asbestos-impregnated sheets shall then be corrugated and fabricated into pipe sections in accordance with the requirements for class I pipe. After fabrication, the pipe shall be coated inside and outside with bituminous material as specified in 3.6.1.

3.6.8 Coating H. Pipe shall be asbestos-impregnated and bituminous coated as specified in 3.6.7, and shall have a smooth paved invert as required in 3.6.2.

3.6.9 Coating J. Pipe shall be asbestos-impregnated and bituminous coated as specified in 3.6.7, and shall be fully paved as specified in 3.6.5.

3.6.10 Coated pipe bands. Pipe bands furnished for coated pipe shall be uniformly coated with bituminous material inside and outside to a minimum thickness of 0.05 of an inch measured on the crests of corrugations for corrugated bands and on the surfaces of other types of bands (see 4.4.5.2).

3.6.11 Coating P. Mill applied pitch resin or polymer coatings are applied to one or both sides of standard ASTM A 444 galvanized culvert sheet modified to include sheet in thicknesses 0.040 inch, 0.034 inch, and 0.028 inch.

Each 2 to 5 feet (0.61 to 1.52 M) of precoated sheet in coils or cut lengths shall be identified as follows:

1. Name of Sheet Manufacturer
2. Brand Name
3. Specified Thickness Steel
4. Type of Precoated Sheet
5. Identification Symbols Relating To A Specifiec Heat Number, Lot Number, and Coating Number
6. Federal Specification Number (WW-P-405)

All markings shall be placed on the bare or thinner coating side, unless otherwise specified.

3.6.11.1 Non-metallic coating types.

Type A - Sheet coated with pitch resin or polymer in a thickness of 0.010 inch minimum one surface only.
Type B - Sheet coated with pitch resin or polymer in a thickness of 0.010 inch minimum on one surface and polymer in a thickness of 0.0003 inch minimum on the other side.
Type C - Sheet coated with pitch resin or polymer in a thickness of 0.010 inch minimum on both sides.

3.6.11.2 Commercial practice. The heavier coating specified above is normally on the inside of the fabricated pipe but may be placed on the outside, if the most severe corrosion or abrasion is expected there. All tests included in the specification are performed on the side with the heavier coating thickness.

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3.6.11.3 Non-metallic coating. The pitch resin or polymer coating shall conform to the specifications listed in table IX and shall be capable of being applied to the material specified in 3.1.1, the cured coating shall be sufficiently flexible so that it will withstand corrugating and culvert forming, and lock seaming operations.

3.7 Workmanship. Pipe shall show careful finished workmanship in all particulars and shall be free from the following defects:

(a) Undue deviation from true shape.
(b) Uneven laps.
(c) Variations from a reasonably true centerline.
(d) Ragged or diagonally sheared edges.
(e) Loose, unevenly lined or spaced rivets, bolts, or welds.
(f) Poorly formed rivet heads.
(g) Illegible brand.
(h) Bruised, scaled, or broken zinc coating.
(i) Dents or bends, other than corrugations.
(j) Spot welds - crack, tip pickup, pits, and metal expulsion.
(k) Manually deposited arc welds - cracks, and closely spaced in-line surface porosity.
(l) Spiral machine welds - cracks, skips, or deficient welds.
(m) Corroded or improperly cleaned and painted welds.
(n) Poorly formed lock seams and damaged lock seam metal.
(o) Inadequate, improperly applied, cracked, or loose asphalt coating.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. The sample size levels of inspection shall be in accordance with MIL-STD-105.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all pipe and structural sections of the same class, shape, and coating offered for delivery at the same time.

4.2.2 Sampling for dimensional and visual examination of end item. Sampling of corrugated pipe for these characteristics shall be in accordance with inspection level S-4.

4.2.3 Sampling for tests. Samples for tests shall be selected in accordance with inspection level S-2.

4.3 Examination. The acceptable qualith levels (AQL's) shall be in accordance with MIL-STD-105.

4.3.1 End item. Pipe shall be examined in accordance with table VIII. The AQL's for defects listed in table VIII shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

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TABLE VIII. Classification of defects, end item

Defects	Major	Minor
Class, shape, and coating not as order or specified.....	X	
Dimensions including diameter, length, and thickness of metal not as specified.....	X	
Uneven laps.....		X
Deviation from a true center line.....		X
Ragged or diagonally sheared edges.....		X
Loose, unevenly lined or spaced rivets or bolts.....	X	
Poorly formed rivet heads.....	X	
Illegible brand marking.....		X
Dents or bends, other than corrugations.....		X
Cracks or metal expulsion in spot welds.....	X	
Tip pickup or pits in spot welds.....		X
Cracks or closely spaced in-line surface porosity in manually deposited arc welds.....	X	
Cracks, skips, or deficient welds in spiral machine welding..	X	
Welds corroded or not painted.....		X
Lock seams poorly formed or damaged.....	X	
Asphalt coating or paving significantly cracked or loose.....	X	

4.3.1.1 Pitch of corrugations. Measure pitch at right angles to the direction of corrugations.

4.3.1.2 Thickness of sheet and plate. Measure thickness normal to the tangent of corrugation.

4.3.1.3 Measure all dimensions of classes I and II pipe arches from the inside crest of corrugations.

4.3.1.4 Measure dimension "B" in table VI of classes I and II pipe arches vertically from the lowest portion of the pipe arch base to a horizontal line drawn across the widest part of the arch.

4.4 Test methods.

4.4.1 Tension tests and chemical composition, classes I, II and III.

4.4.1.1 Tension tests. Two tension tests shall be made from each heat and shall be tested in accordance with the provisions of ASTM A 370 (see 3.1.1 and 3.5.1).

4.4.1.2 Chemical compostion, base metal. Perform check analysis for chemical composition in accordance with method 111 or 112 of Fed. Test Method Std. No. 151. In case of dispute, method 111 shall be used for reference analysis (see 3.1.1 and 3.5.1).

4.4.2 Weight of zinc coating.

4.4.2.1 Classes I and II. Determine weight of zinc in accordance with any of the applicable procedures in ASTM A 90 (see 3.1.1 and 3.2.4).

4.4.2.2 Class III. Determine weight of zinc in accordance with AASHO M 167 (see 3.2.4 and 3.5.2).

4.4.3 Fasteners. Conduct tensile, bending, and hardness tests in accordance with ASTM A 370 (see 3.5.3).

4.4.4 Spot welds.

4.4.4.1 Specimens. Prepare a total of two specimens for qualifying welds and for welding procedure approval. Lap two 1-1/2 inch minimum width by 5 inch length strips of corrugated culvert sheet, representative of the material to be welded, and join them together by a single spot weld. The size of the spot weld and the width of lap should be the same as those used in production. The longer axis of the specimen should be parallel to the direction of rolling. Weld the specimens after the resistance spot welding unit has been properly adjusted for the sheet gage and for each brand or type of steel.

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4.4.4.2 Procedure. Mount each specimen in turn on a standard calibrated tensile testing machine and test to destruction in tension (see 3.3.3.3 and table IV).

4.4.4.3 Frequency of test. Perform tests at each change in machine adjustment, operators, gage, or brand of steel being fabricated.

4.4.5 Coatings, bituminous.

4.4.5.1 Coatings A to J. Conduct tests in accordance with AASHO M 190 (see 3.6) except for coating F.

4.4.5.2 Coatings A to J. Measure thickness of the coating at an exposed end or with a magnetic or scratch gage (see 3.6.1, 3.6.2, 3.6.6, 3.6.9 and 3.6.10).

4.4.5.3 Coating F.

4.4.5.3.1 Fillers.

(a) Weight. Determine percent of wet weight in accordance with ASTM D 1856 or method 4021.1 in Fed. Test Method Std. No. 141. Use benzol and extract at least five times when using method 4021.1 (see table VII).

(b) Fineness. Determine fineness of fillers in accordance with ASTM D 147 (see 3.6.6.8.2).

4.4.5.3.2 Asphalts. Determine the melting point and the penetration of the asphalt extracted by the Abson method in ASTM D 1856 (see 3.6.6.3.1).

4.4.5.3.3 Volatiles. Determine percent by weight in accordance with ASTM D 1203 or method 4041.1 in Fed. Test Method Std. No. 141 (see table VII).

4.4.5.3.4 Nonvolatile bitumens. Calculate weight by difference in accordance with method 4053 in Fed. Test Method Std. No. 141 (see table VII).

4.4.5.3.5 Pigment. Determine percent by volume of nonvolatile pigment in accordance with ASTM D 1010 or method 4311 in Fed. Test Method Std. No. 141 (see table VII).

4.4.5.3.6 Weight per gallon. Determine weight in pounds in accordance with ASTM D 1010 or method 4184 in Fed. Test Method Std. No. 141 (see table VII).

4.4.5.3.7 Flash point of solvents. Determine flash point temperature in accordance with ASTM D 93 or method 4293 in Fed. Test Method Std. No. 141 (see table VII).

4.4.5.3.8 Drying time. Use a steel sheet or plate of any convenient size, or a section of pipe. Apply sufficient wet film coating to develop a 1/16 inch dry film in an environment at 75° ± 2°F. and 50 ± 5 percent relative humidity. Touch the coating after it has set 4 hours from the start of drying. Allow the coating to dry for a total period of at least 36 hours. Apply pressure using thumb and fingers. Examine fingers to determine whether any coating adheres. A tacky condition shall not be cause for rejection (see 3.6.6.3).

4.4.5.3.9 Working properties. Spray the mastic on a sample of pipe or a steel sheet or plate of any convenient size. Temperature of the mastic should be 60° ± 10°F. Tank pressure should be not more than 80 p.s.i.g. and the air pressure at the gun nozzle should be no more than 80 p.s.i.g. Examine the sprayed surface (see 3.6.6.5).

4.4.5.3.10 Sag. Apply a 1/8 inch wet film coating of the mastic down hand on a steel sheet or plate of an convenient size. Suspend the sprayed specimen vertically for one hour immediately after spraying (see 3.6.6.7).

4.4.5.4 Coating P. Hill applied pitch resin or polymer type. Determine the properties of the pitch resin or polymer coating in accordance with table IX.

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TABLE IX. Specifications on Pitch-resin or polymer coating

Property	Test Method	Specifications
Adhesion	-----	See para. 4.4.5.4.1
Thickness, mils	ASTM D 1005	10 minimum
Impact 1/, inch pounds	Gardner	35
Freedom from holidays	-----	See para. 4.4.5.4.2
Imperviousness	-----	See para. 4.4.5.4.3
Freeze thaw	-----	See para. 4.4.5.4.4
Tensile strength 2/, psi	ASTM D 638	1500 minimum
Elongation 2/, %	ASTM D 638	300 minimum
Tear resistance, ppi	ASTM D 1004	250 minimum
Hardness, shore D (10 sec)	ASTM D 2240	30 minimum
Dielectric strength, volts/mil	ASTM D 149	350 minimum
Water absorption, weight % in 24 hours	ASTM D 570	0.10 maximum
Resistance to acid solution 3/	ASTM D 543	No detrimental effect
Resistance to basic solution 4/	ASTM D 543	No detrimental effect
Resistance to salt solution 5/	ASTM D 543	No detrimental effect
Weatherability 6/	ASTM D 529	1000 hours
Resistance to Microbial Attack 7/	ASTM G 22	No attack
Flammability	ASTM D 568	Non burning or self-extinguishing
Abrasion resistance, coefficient	See footnote 8/	
Adhesive peel strength 9/, ppi	ASTM D 903	
@ 73° F		40 minimum
-40° F		10 minimum
145° F		25 minimum

NOTES 1/ Coating bonded to 16 gage galvanized steel, impacted with gardner variable impactor with 5/8 inch punch and with specimen set on 0.640 inch punch die. The test should be performed at 77° F. (With a 20 milcoating thickness, the impact should exceed 100 inch pounds.)
2/ Specimens pulled at two inches per minute.
3/ Sulfuric acid, 30 percent.
4/ Sodium hydrozide, 10 percent.
5/ Sodium chloride, 10 percent.
6/ 2 hour cycle including 18 minutes of waterspray at a top temperature of 140° F.
7/ Use procedure B.
8/ The abrasion resistance shall be determined by a grit blast method. With a modified "spark plug tester" using CL-2 grit and 100 psi air pressure, the coating shall abrade at a rate of less than 0.1 mils per second. Note: This is an interim method. As soon as data is available, specifications shall be established using test method ASTM D 658.
9/ 28 gage galvanized steel bonded to 16 gage galvanized steel at 350° F for 30 seconds, specimens pulled at 10 inches per minute.

4.4.5.4.1 Adhesion. A coupon 2" x 3" shall be cut from the sample of precoated steel. The coupon shall be bent 180 degrees over a 0.5 inch mandrel (surface with the thicker pitch-resin or polymer coating shall be on the outside of the bend). There shall be no spalling or cracking of the coatings. A cut through the pitch-resin or polymer coating shall be made across the outside radius of the bend. There shall be no detachment of the coating at the cut. This test shall be performed at 0° F, 77° F, and 120° F (± 2° F).

4.4.5.4.2 Holidays. The pitch-resin or polymer coating on the steel shall be free of holidays in the thicker coating. Pinholes or voids shall be detected by a high voltage holiday detector using 100 volts per mil of coating thickness.

4.4.5.4.3 Imperviousness. The pitch-resin or polymer coating shall be tested for imperviousness to chemical reagents according to the applicable sections of ASTM D 543. The reagents used shall be as described in sections 4.4.40, 4.4.42, and 4.4.46 or ASTM D 543. The test procedure shall be modified as follows: The reagents shall be held in separate valleys of the coated corrugations for a period of 48 hours during which time no loosening or separation of the pitch-resin or polymer shall occur.

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4.4.5.4.4 Freeze thaw. Three coupons 6" x 6", shall be cut from the sample of precoated metal. The coupons shall be immersed in water at ambient temperature for two weeks and then be subjected to an extended period of freeze/thaw cycling. The cycle shall consist of 8 hours at 0° F and 16 hours at room temperature. Cycle time may be extended over weekends and holidays but such variation should be noted. Observations and testing of one of the samples shall be made after 35 cycles to determine brittleness and susceptibility to chipping when cold, behavior on bending around a 0.5 inch mandrel, spalling on freezing and thawing, maintaining of bond to the steel, or other apparent changes in coating materials as a result of freezing and thawing. These observations and tests shall be conducted again after 70 and 100 cycles. Coating shall withstand 100 cycles without detrimental effects.

4.4.5.4.5 Test for adhesion, thickness, impact and holidays are intended to insure that the product meets the requirements of the specification in continous mass production and are applicable for process control or product acceptance.

4.5 Rejection

4.5.1 Tests. A lot shall be rejected for failure to meet the test requirements of this specification when tested in accordance with 4.4.

4.5.2 Retests. Conduct retests using at least 2 specimens.

4.6 Inspection of preparation for delivery. The packaging, packing, and marking of the pipe shall be inspected to determine conformance with the applicable requirements of Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Bolts. Bolts or other fastening devices shall be packaged in quantities as specified in close-fitting boxes conforming to PPP-B-636 class weather-resistant. Close, waterproof seal and strap the boxes in accordance with the appendix to PPP-B-636.

5.1.2 Level C.

5.1.2.1 Bolts. Bolts or other fastening devices shall be packaged in accordance with the suppliers' standard practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Class I and Class II pipe. Pipe up to 24 inches in diameter shall be bundled by placing a layer on two 4 x 4 x 48 inch timbers and strapping with type I, class B flat steel strapping conforming to QQ-S-781. Strapping shall be secured to anchor straps nailed to each end of the timbers, drawn tight and sealed. Appropriate chocks shall be nailed on the ends of the timbers to prevent movement of the pipe and to facilitate strapping. Straps and separators shall be placed approximately 1/16 of the length of the bundle from each end. Each like bundle containing a single layer of pipe shall be capable of being stacked and bound to each other. Pipe above 24 inches in diameter shall be shipped loose.

5.2.1.2 Class III pipe. Pipe shall be bundled using type I, class B strapping conforming to QQ-S-781, utilizing corner protectors and lifting plates.

5.2.1.3 Bolts and coupling bands. Bolts packaged in accordance with 5.1 and coupling bands shall be packed in quantities as specified in close-fitting boxes conforming to PPP-B-601, overseas type.

5.2.2 Level B.

5.2.2.1 Pipe, bolts, and coupling bands. Pipe, bolts, and coupling bands shall be packed as specified for Level A except that boxes shall be domestic type and strapping need not be class B.

5.2.3 Level C.

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5.2.3.1 Pipe, bolts, and coupling bands. Pipe, bolts, and coupling bands shall be packaged in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Markings (see 6.2).

5.3.1 Civil agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Pipe and structures covered by this specification are intended for use as culverts, storm drains, underdrains, conduit, and miscellaneous applications. Where additional protection is required because of corrosive or erosive conditions a coating type should be specified for both pipe and coupling bands.

6.1.1 Mill applied pitch resin or polymer coatings extend the life of corrugated steel pipe and are intended for installations requiring longer maintenance free operation than is available from plain galvanized pipe. The coatings have superior adhesion to a galvanized surface for all temperatures, are self-extinguishing in fire, have properties which do not change significantly with age and more highly corrosion and abrasion resistant than the asphalt based coatings of section 6. (See para. 3.6.11).

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class, series, and shape (see 1.2.1).
(c) Coating (see 1.2.2 and 6.3).
(d) Classes I and II, perforations for drainage, and pattern (see 3.1.2 and table I).
(e) Classes I, II, and III, metal thickness (see 3.2.2 and table III).
(f) Classes I, II, and III, number, length, and diameter of sections, and total length required (see 3.2.5 and 6.4).
(g) Classes I and II, kind of longitudinal or helical seams, if one is preferred (see 3.3.1 and 3.4.2).
(h) Class I and II bands, class, type, number and whether one or two-piece construction (see 3.3.4.1).
(i) Shape 3, all classes, span and rise required (see 3.3.6.3, 3.4.5.3, and 3.5.5.3).
(j) Class II, shape, and when smooth-lined pipe is required (see 3.4.1).
(k) Other fastening bands, if required (see 3.4.3.2).
(l) Class III, number of bolts per foot of longitudinal seams, if more than four per foot are required (see 3.5.3).
(m) Class III, shape 4, dimensions, span, and rise required and bearing for arches (see 3.5.5.4.1 and 3.5.5.4.2).
(n) Size of lot if different from 4.2.1.
(o) Selection of applicable level of packaging (see 5.1).
(p) Selection of applicable level of packing (see 5.2).
(q) Whether civil or military marking is required (see 5.3).
(r) Class 1, coupling band gaskets (see 6.5).
(s) Sheet steel manufacturer's analysis (see 6.6).

6.3 Coatings.

6.3.1 Coating A to P. Coatings are available for the following classes of pipe:

Coatings	Classes
A	I and II
B	I and II
C	I and II
D	I and II
E	I and II
G	I
H	I
J	I
F	III
P	I and II

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6.3.2 Coating F. This coating is intended for application by spraying but brushing or troweling may be used for small areas or where spraying is not practicable. It should be applied as one thick coat to obtain a minimum dry film thickness of 0.05 inch. High pressure spray equipment is necessary for spray application.

6.4 Length. The length of pipe and structural sections should be subject to modification when necessary to conform to transportation regulations.

6.5 Coupling bands. Rubber gaskets for installation under metal coupling band gaskets may be used for installations where it is necessary to prevent infiltration of fill material that consists of fine-grained noncohesive soil.

6.6 Steel sheet manufacturer's analysis.

6.6.1 Classes I and II.

6.6.1.1 Accepted brands of metal. No metal should be accpeted under these specifications and no bids should be considered for the materials described in 6.2 until after the sheet manufacturer's certified analysis and manufacturer's quarantee have been passed upon by the purchaser and accepted. Misbranding or other misrepresentation will each be considered a sufficient reason to discontinue the acceptance of any brand under these specifications and notice sent to the sheet manufacturer of the discontinuanace of acceptance of any brand will be considered to be notice to all culvert companies which handle that particular brand. One brand, and one brand only should be approved for each kind of base metal furnished by each of the actual manufacturers of the sheets.

6.6.1.2 Sheet manufacturer's certified analysis. The manufacturer of sheet indentified by brand or mark should file with the purchaser a certificate setting forth the name or brand of metal to be furnished, the specified chemical composition, and a typical or average analysis showing the percent carbon, manganese, phosphorus, sulfur, silicon, copper, and any other elements specified for the particular kind of base metal. The certificate shall be sworn to for the manufacturer by a person having legal authority to bind the company.

6.6.1.3 Sheet manufacturer's guarantee. The manufacturer of the sheets should submit with the certified analysis a guarantee providing that all metal furnished shall conform to the specification requirements, should bear a suitable indentification brand or mark, and should be replaced without cost to the purchaser when not in conformity with the specified analysis, sheet thickness or spelter coating, the cost to be limited to the replacement of the material only. The guarantee should be so worded as to remain in effect so long as the manufacturer continues to furnish the material.

6.6.2 Class III. Mill test reports. Mill test reports of chemical composition by ladle analysis and mechanical properties should be on file with the manufacturer. Test reports of weight of zinc coating should be on file with the fabricator. If requested, test reports of chemical composition, mechanical porperties, and weight of zinc coating may be obtained for heat and coating lots selected at random by the purchaser.

6.7 Design considerations. For information or design of culverts, consult the following:

Corrugated Metal Pipe Culverts,
Structural Design Criteria and
Recommended Installation Practices,
U.S. Department of Commerce,
Bureau of Public Roads.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Handbook of Steel Drainage and
Highway Construction Products

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, NY 10017.)

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MILITARY CUSTODIANS:

Army - ME
Navy - YD
Air Force - 82

Reivew activities:

CE, CS

Military Coordinating Activity:

Army - ME

Project No. 4710-0325

Preparing activity:

DOT-FHW

Civil Agency Coordinating Activities:

GSA-FSS
HEW-FEC
INTERIOR-BPA
USDA-AFS

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